



UNIVERSIDADE D
COIMBRA

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NAVIGATING MODAL SHIFT: EXPLORING PUBLIC TRANSPORT PREFERENCES IN COIMBRA

VOLUME 1

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PREVIEW

TABLE OF CONTENTS

CHAPTER 1: INTRODUCTION	1
1.1 MOTIVATION AND PROBLEM OVERVIEW	1
1.2 RESEARCH OBJECTIVES AND PLAN	4
1.3 DISSERTATION STRUCTURE	4
1.4 RESEARCH GAP AND RELEVANCE	5
CHAPTER 2: LITERATURE REVIEW	7
2.1 SUSTAINABLE MOBILITY – CAR DEPENDENCY AND MITIGATION MEASURES	7
2.2 PUBLIC TRANSPORT	10
2.3 MODAL SHIFT	12
2.4 QUALITY ATTRIBUTES THAT IMPACT PUBLIC TRANSPORT DEMAND	13
2.4.1 PUBLIC TRANSPORT QUALITY ATTRIBUTES	13
2.4.2 OBJECTIVE ATTRIBUTES	14
2.4.3 PERCEIVED ATTRIBUTES	22
2.5 CONCLUSION OF THE LITERATURE REVIEW	26
CHAPTER 3: METHODOLOGY	27
3.1 SURVEY	27
3.1.1 SURVEY DEVELOPMENT	27
3.1.2 SURVEY SEGREGATION	28
3.1.3 SAMPLE DESIGN AND RESPONDENT SELECTION PROCEDURE	29
3.1.4 INCLUSION AND EXCLUSION CRITERIA	30
3.2 DATA ANALYSIS	30
CHAPTER 4: CASE STUDY, COIMBRA	33
4.1 STUDY AREA: COIMBRA	33
4.2 PUBLIC TRANSPORT IN COIMBRA	34
4.3 INTRODUCTION OF THE METRO MONDEGO PROJECT AND EXPECTATIONS	35
CHAPTER 5: RESULTS	38
5.2 DESCRIPTIVE ANALYSIS	38
1 ST SECTION: PERSONAL CHARACTERISTICS	38
2 ND SECTION: MOBILITY BEHAVIOUR PATTERNS	42
3 RD SECTION: PRIVATE CAR USERS	47
4 TH SECTION: PUBLIC TRANSPORT USERS	53
5 TH SECTION: VALUED ATTRIBUTES IN PUBLIC TRANSPORT	57
CHAPTER 6: DISCUSSION	63

1ST SECTION: PERSONAL CHARACTERISTICS	63
2ND SECTION: MOBILITY PATTERNS	63
3RD SECTION: CAR USERS	64
4TH SECTION: PUBLIC TRANSPORT USERS	67
5TH SECTION: VALUED ATTRIBUTES IN PUBLIC TRANSPORT	68
ENDING QUESTIONS	72
 CHAPTER 7: CONCLUSION	 74
 STUDY LIMITATIONS AND FUTURE RESEARCH	 75
 REFERENCE LIST	 76
 APPENDIX	 90
 APPENDIX A: PUBLIC TRANSPORT SATISFACTION SURVE	 90

List of Figures

Figure 1. Global GHG emissions by sector (left axis, bars) and per capita (right axis, black line), 1970-2022 (in Gt CO ₂ eq/ku). (Crippa et al., 2023).	2
Figure 2. Greenhouse gas emissions from transport in Europe. (Eurostat, 2023)	2
Figure 3. Modal Split of Land Passenger Transport, 2014 (% of total land passenger-kilometers). (Eurostat, 2017).	3
Figure 4. Map of Coimbra Municipality. (Turismo Centro Portugal, 2020).	33
Figure 5. SMUTUC Network Map. (CMC, 2024).	34
Figure 6. Metro bus rapid transit system. (Metro Mondego, 2024).	35
Figure 7. Metro Mondego Network. (Metro Mondego, 2024).	36
Figure 8. Metro Mondego Main Advantages. (Lopes, 2024).	36
Figure 9. Number of survey Respondents by Age Group.	38
Figure 10. Number of Respondents by Gender.	39
Figure 11. Number of Respondents by the highest Level of Education Completed.	40
Figure 12. Number of Respondents by Employment Status.	40
Figure 13. Respondents' Regular Commuting Trips to/from Coimbra Municipality.	41
Figure 14. Vehicle Ownership.	42
Figure 15. Daily Commute Time.	43
Figure 16. Frequency of Public Transport Use.	44
Figure 17. Frequency of Private Vehicle Use.	44
Figure 18. Distance from Home to Nearest Public Transport Stop.	45
Figure 19. Primary mobility option within Coimbra municipality.	46
Figure 20. Main reasons for choosing to use a car instead of Public Transportation.	47
Figure 21. Factors that could make public transportation more attractive to respondents.	48
Figure 22. Possibility of Switching to Public Transportation as an Alternative to Using a Car.	49
Figure 23. Respondents' Preferred Commute Alternatives in Response to High Fuel or Elevated Parking Prices.	50
Figure 24. Main reasons for choosing Public Transportation.	53
Figure 25. Main Challenges/Drawbacks Faced When Using Public Transportation.	54
Figure 26. Improvements to Encourage More Frequent Use of Public Transport.	55

Figure 27. Assessment of the 6 quality attributes (accessibility, price, frequency, punctuality, integration, duration) of Category 1 (Service) on an importance scale of 1 to 5. _____	58
Figure 28. Assessment of the 5 quality attributes (maintenance, parking, bicycles, inclusivity, sustainability) of Category 2 (Additional Services) on an importance scale of 1 to 5. _____	59
Figure 29. Assessment of the 2 quality attributes (on-board safety, stop safety) of Category 3 (Safety) on an importance scale of 1 to 5. _____	60
Figure 30. Assessment of the 3 quality attributes (cleanliness, overcrowding, comfort) of Category 4 (Comfort) on an importance scale of 1 to 5. _____	61
Figure 31. Assessment of the 2 quality attributes (information, complaints) of Category 5 (Information and Customer Service) on an importance scale of 1 to 5. _____	62

List of Tables

Table 1. Public Transport (PT) Quality Attributes.	14
Table 2. Inclusion and Exclusion Criteria for the Customer Satisfaction Survey.	30
Table 3. Sample characteristics.	42

“The role of transport policy is not to eradicate the car, but rather to establish high-quality alternatives to eliminate the need for one” (Van Exel & Rietveld, 2009)

Abstract

Transport emissions account for almost a quarter of all greenhouse gas (GHG) emitted in Europe. As such, there is a growing effort to support modal shift strategies that can significantly reduce emissions in the transport sector. This dissertation investigates customer satisfaction (CS) with public transport (PT) services in Coimbra, focusing on the factors influencing both current PT users and private car users within the Department of Mechanical Engineering at the University of Coimbra. This study analyzed car and PT users to identify the key quality attributes of PT services that attract current users and determine what changes could encourage car users to switch to PT. A comprehensive customer satisfaction survey (CSS) was conducted, revealing that improvements in transport frequency, accessibility, reliability, information provision, and comfort are essential to increasing PT usage. These insights contribute to the broader understanding of how to effectively promote a shift towards more sustainable transport options in urban environments.

Keywords: Public transport, car use, modal shift, customer satisfaction, quality attributes, service quality, survey

Resumo

As emissões de gases de efeito de estufa (GEE) produzidas pelo setor dos transportes representam quase um quarto de todos os GEE emitidos na Europa. Como tal, há um esforço crescente para apoiar estratégias de mudança modal que possam reduzir significativamente as emissões no setor dos transportes. Esta dissertação investiga a satisfação dos clientes com os serviços de transporte público (TP) em Coimbra, com foco nos fatores que influenciam tanto os atuais utilizadores de TP quanto os utilizadores de automóveis privados no Departamento de Engenharia Mecânica da Universidade de Coimbra. Este estudo analisou utilizadores de automóveis e de TP para identificar os principais atributos de qualidade dos serviços de TP que atraem os utilizadores atuais e determinar que mudanças poderiam incentivar os utilizadores de automóveis a mudar para o TP. Foi realizado um inquérito abrangente de satisfação do cliente, que revelou que melhorias na frequência dos transportes, acessibilidade, fiabilidade, fornecimento de informação e conforto são essenciais para aumentar o uso do TP. Estas perceções contribuem para uma compreensão mais ampla de como promover eficazmente a mudança para opções de transporte mais sustentáveis em ambientes urbanos.

Palavras-chave: Transporte público, uso de automóvel, mudança modal, satisfação do cliente, atributos de qualidade, qualidade do serviço, inquérito

List of Abbreviations

COVID-19	Coronavirus Disease-19
CS	Customer Satisfaction
CSS	Customer Satisfaction Survey
EC2U	European Campus of City-Universities
EU	European Union
GHG	Greenhouse Gas
INE	Instituto Nacional de Estadística
ITS	Intelligent Transport System
SDGs	Sustainable Development Goals
SEM	Structural Equation Modelling
SQ	Service Quality
PSQ	Perceived Service Quality
PT	Public Transport
PT-SQ	Public Transport Service Quality

PREVIEW

Chapter 1: Introduction

1.1 Motivation and Problem Overview

Climate change is one of the most critical challenges to human society. To combat it, global action is underway. In 2015, the Paris Agreement united 196 countries in the fight against climate change (UNFCCC, 2024). Europe, the USA, Canada, Brazil, and Australia have pledged to achieve net zero emissions by 2050, while China and Saudi Arabia are targeting 2060 and India aims for climate neutrality by 2070. The EU's ambitious targets outlined in the European Green Deal and European Climate Law aim to reduce net domestic greenhouse gas (GHG) emissions by at least 55% by 2030, compared to 1990 levels (Crippa et al., 2023). Furthermore, in 2015, for the first time, all 193 United Nations members approved an international plan, the 2030 Agenda, to end poverty, protect the planet, and improve quality of life. The 2030 Agenda is a 15-year plan, created based on 5 Principles – People, Planet, Prosperity, Partnership, and Peace - that encompasses 17 Sustainable Development Goals (SDGs), 169 targets, and over 230 universal indicators that allow monitoring of the progress made in each target (United Nations 2022).

Despite worldwide recognition, global GHG emissions have been on the rise, with the transport sector contributing significantly. In 2019, global GHG emissions peaked with a 23% contribution from the transport sector (Jaramillo et al., 2023). The coronavirus disease-19 (COVID-19) pandemic led to a temporary decrease in emissions in 2020 (Figure 1 and Figure 2); however, post-pandemic levels have surpassed previous records. In 2022, global GHG emissions increased by 6.2% compared to 2020 with the major contributors being China, USA, India, European Union (EU), Russia, and Brazil, collectively accounting for over half of the global population, Gross Domestic Product, fossil fuel consumption, and GHG emissions (Crippa et al., 2023).

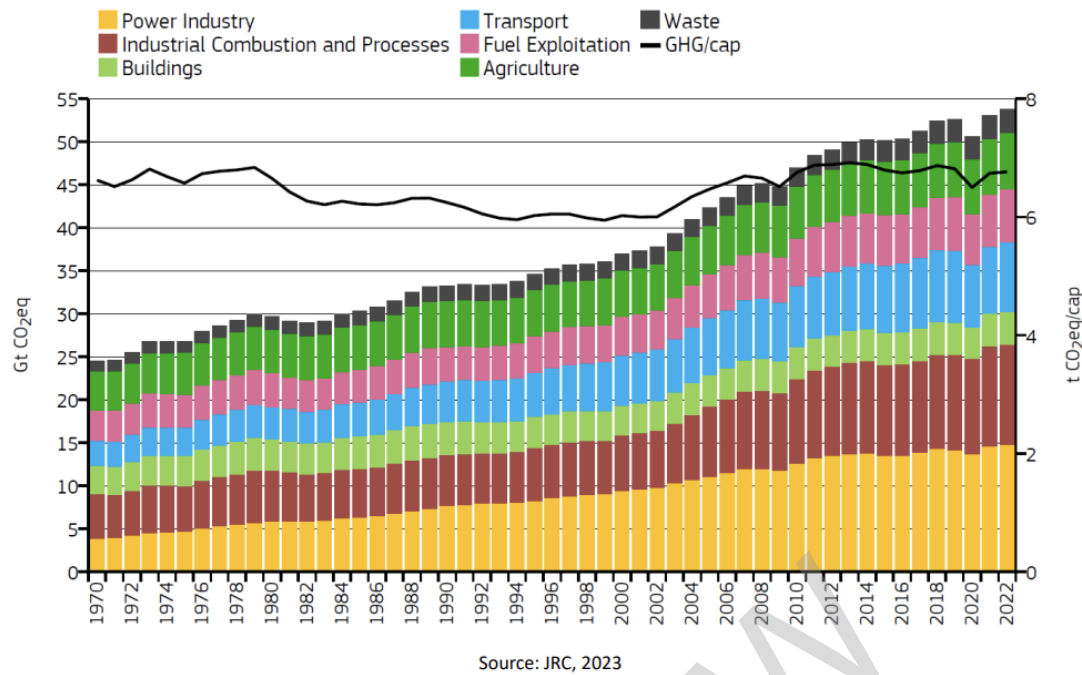


Figure 1. Global GHG emissions by sector (left axis, bars) and per capita (right axis, black line), 1970-2022 (in Gt CO₂eq). (Crippa et al., 2023).

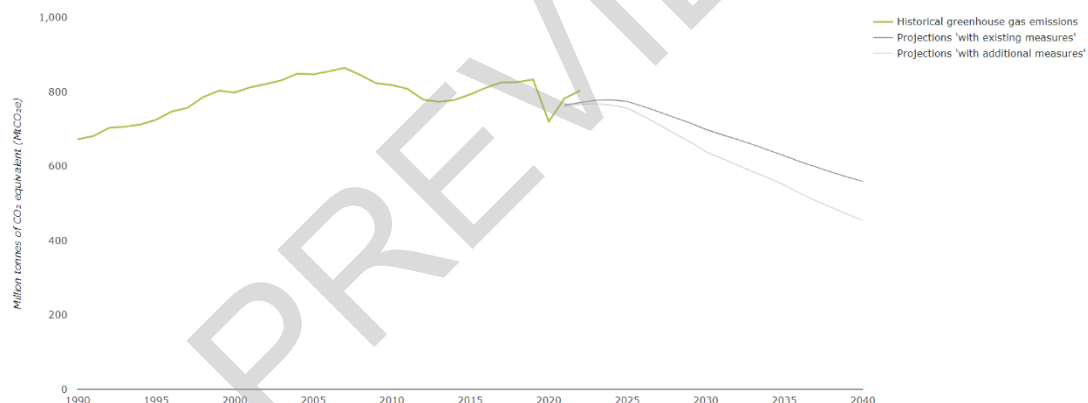


Figure 2. Greenhouse gas emissions from transport in Europe. (Eurostat, 2023)

This context underscores the urgency to reduce overall GHG emissions, particularly in the transport sector. Curbing emissions and meeting climate targets will require adopting sustainable mobility options and modal shift strategies. This entails transitioning away from ownership-based models and pollutant transport options towards efficient public transport (PT) services and active mobility (e.g., walking and biking).

Despite its significant importance, PT systems are not the preferred choice of transport in Europe (Eurostat, 2023), with Portugal serving as a prime example of this trend (Figure 3).

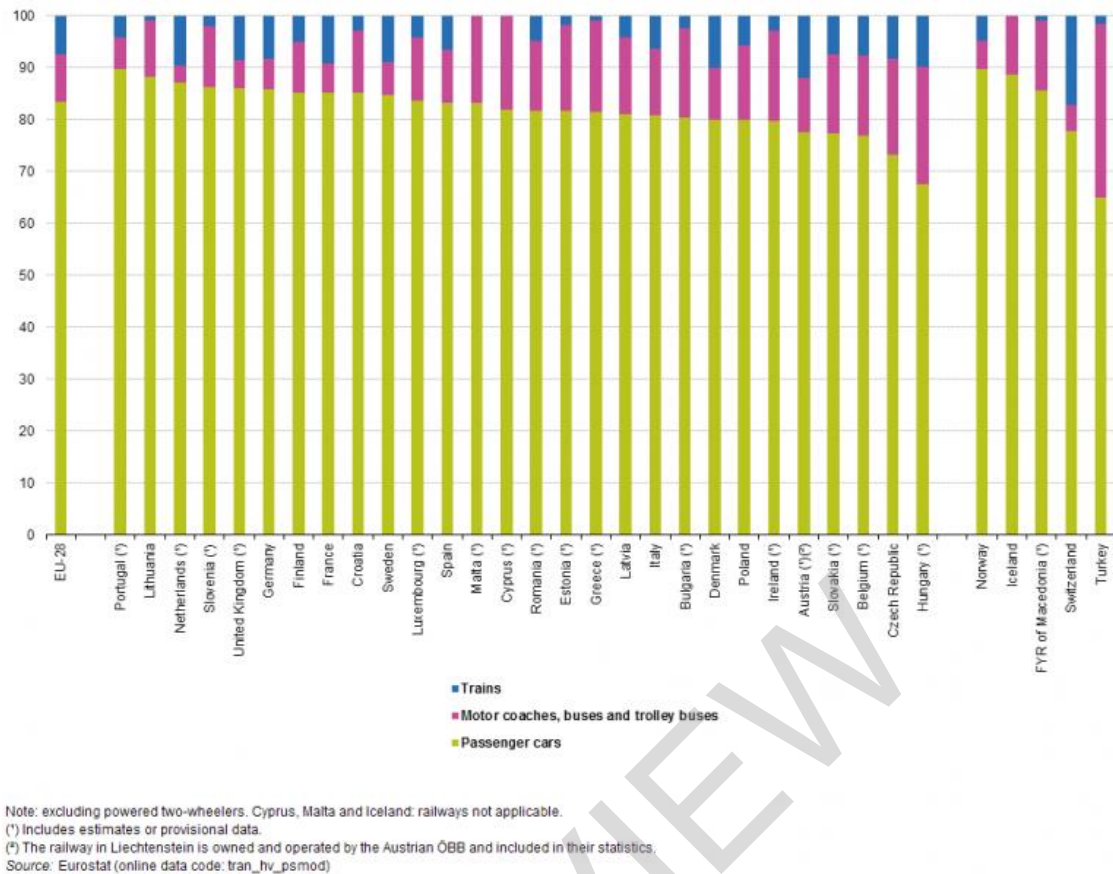


Figure 1. Modal Split of Land Passenger Transport, 2014 (% of total land passenger-kilometers). (Eurostat, 2017).

As demonstrated in Figure 3, the citizens of Portugal often favor private car usage instead of bus or train services (Eurostat, 2017). Understanding this decision is imperative for effective mitigation efforts on car use externalities. Hence, this dissertation sets out to provide further insight into the customer perception of the general population in Coimbra regarding PT use and to identify the factors influencing their modal choices.

This dissertation is part of an innovative international initiative focused on educating students about sustainable practices and urban policies essential for the development of resilient cities capable of confronting future climate challenges. A part of this initiative is the Master in Sustainable Cities and Communities, coordinated by the Faculty of Science and Technology at the University of Coimbra, and funded by the European Campus of City-Universities (EC2U) project.

1.2 Research Objectives and Plan

To efficiently plan a PT system it is crucial to understand which quality attributes influence modal choice. The main goal of this study is to understand the changes required in the PT system to motivate a modal shift in Coimbra from car use to PT; and understand how to improve perceived service quality (PSQ) to maintain current PT users. The associated research questions, based on the relevant research review by Redman et al. (2013), can be stated as follows:

- (1) What quality attributes of PT services attract current users?
- (2) What changes in quality attributes of PT services would encourage a modal shift from private automobile users to PT?

The research plan, based on similar satisfaction studies, follows four main steps, which will be further explained in the Methodology section:

1. Survey development
2. Survey distribution within the Department of Mechanical Engineering from the University of Coimbra
3. Descriptive analysis of the obtained results
4. Discussion and resume of the main conclusions of the survey

1.3 Dissertation Structure

The present dissertation is organized into seven Chapters. Below is a brief resume of each chapter's content.

Chapter 1 is an introductory chapter that presents the reader to the study's problem, research gap, and relevance. Furthermore, this section reveals the main research objectives, selected methodology, and overall document organization.

Chapter 2 will present the primary findings of a qualitative and unstructured literature review on sustainable mobility, public transportation, modal shift dynamics, and factors influencing mode choice. Subsequently, a concise sub-chapter outlines the key quality attributes of PT identified in the literature review. To conduct this review, an online search in the Web of Science database was developed to identify specific studies performed in Portugal and understand the research gap. From this search, 12 results were

obtained revealing some research on sustainable mobility done in Portugal, mainly with a focus on active mobility (bicycle and pedestrians). Additionally, it revealed a lack of data collection on the modal shift to PT in Portuguese cities. The second phase of this literature review was conducted using the Scopus database. A targeted search strategy was employed, utilizing the following key terms: "sustainable mobility", "public transport", and "modal shift". This initial query yielded 22 potential articles, which were then carefully evaluated based on their titles and abstracts for inclusion in the review. To ensure the analysis reflected current research trends, an additional search was done within the same database, applying a filter to include only articles published within the last five years. This approach allowed the inclusion of recent developments and emerging perspectives in the field of transportation studies. Lastly, the literature base was expanded through the snowball method. This technique involved examining the reference lists of already identified articles, enabling the discovery of additional relevant sources that may have been missed in the initial database searches.

Chapter 3 focuses on the methodology and the selected methods, which are justified based on relevant scientific research with similar objectives to this study. A thorough description of the data collection process (survey) and data analysis is made.

Chapter 4 introduces the case study of Coimbra, its current PT service, and provides further detail on the investments of future PT services, the Metro Mondego Project.

Chapter 5, provides the descriptive analysis of the results obtained in the survey.

Chapter 6 provides a discussion of the results, with possible recommendations and practical suggestions that can improve Coimbra's PT services knowledge on customer satisfaction (CS) and PT ridership.

Chapter 7 presents the reflection of the study's success and the main conclusions gathered, with study limitations and suggestions for future research.

1.4 Research Gap and Relevance

Private vehicles are the preferred transport option in European countries (Figure 3) (Eurostat, 2017). Cars offer unique advantages such as time and spatial flexibility, comfort, and privacy making them the most popular and competitive form of transport (Cruz et al., 2014). On a global scale, this automobile dependency represents urban areas' most significant mobility challenge. Urban populations are already severely affected by

elevated traffic and congestion levels, air pollution, and public space appropriation (Cruz et al., 2014). This is especially concerning due to the rapid rise in urban populations. Currently, half of the world's population lives in urban centers, which is expected to increase to 70% by 2050. Consequently, passenger traffic and vehicle numbers will substantially increase in the next few decades (Fragkias et al., 2013). Thus, it is crucial to create a good quality PT system. However, each region has different mobility needs, hence, to efficiently plan a PT system it is imperative to understand which quality attributes influence the demand (Efthymiou & Antoniou, 2017; Román et al., 2014). This personalized approach is particularly critical given the diverse demographics found within the same population. To meet the expectations of various market segments, PT operators should invest in evidence-based research aimed at enhancing PT services for the entire population by improving PT performance (Abenoza et al., 2017; Beirão & Cabral, 2007). Yet, one of the greatest political barriers to mobility development in urban centers is the difficulty of correct measurement of PT quality, mainly because different populations exhibit different perceptions of PT and, therefore, will have various requirements and expectations (Taylor et al., 2009).

The transport literature has seen a surge in the last decade of scientific research aimed at understanding car dependency and identifying effective strategies to reduce it (Batty et al., 2015; Brand et al., 2013; Gaschi-Uciecha, 2023; Nordfjærn et al., 2014; Redman et al., 2013; Urbanek, 2021a). This focus is particularly critical given that, according to a well-cited systematic review, most studies conducted in this field had revealed weak methodologies and were, often, unsuccessful at curbing car usage (Donald et al., 2014).

Motivated by the aforementioned research gap, this research aims to explore the valued quality attributes that drive PT adherence in Coimbra. Its primary focus is to provide scientific insights to understand how future projects can facilitate a notable modal shift within the city's residents and maintain current ones. Furthermore, this project aims to provide practical information that can be applied by PT providers in Coimbra. To ensure objective results, this study specifically targets car and PT users, thus offering a context-specific understanding of the key factors influencing transportation choices in Coimbra.

Chapter 2: Literature Review

Through a comprehensive review of pertinent scientific literature, this section discusses sustainable mobility, PT, modal shift dynamics, and key quality attributes that influence PT demand. This literature review aims to acquaint the reader with these complex and multifaceted discoveries and provide accurate background information essential for understanding the remaining document.

2.1 Sustainable Mobility – car dependency and mitigation measures

Transportation is a basic and essential requirement in any society that enables individuals to access essential services, resources, and opportunities within their communities, including employment, healthcare, education, and leisure activities (Abenzo et al., 2019). Thus, communities with access to efficient transport options are more financially and socially developed (WBCSD, 2009). However, the transport sector, which encompasses aviation, maritime transport, cars, freight, and PT, significantly contributes to carbon emissions (Jaramillo et al., 2023). Figure 1 highlights the global trend of increasing emissions across different sectors. Despite this trend on a global scale, the EU has achieved a significant reduction of greenhouse gas emissions across various sectors since 1990, but the transport sector remains a notable exception (Figure 2).

The transport sector is responsible for nearly a quarter of total GHG emissions emitted in Europe. Among transport modes, road transport stands out as the largest carbon emitter responsible for 76% of transport emissions in 2021, with private cars producing 60% of those (Ian Tiseo, 2024). Consequently, personal car use emerges as one of the most polluting habits contributing to climate change (Collins & Chambers, 2005; Wall et al., 2007). Furthermore, transportation is a key driver of air pollution and is responsible for the increase in demand for non-renewable resources (Batterbury, 2003; European Environment Agency, 2023a, 2023b; Greene & Wegener, 1997). This is particularly concerning in Portugal, where car dependency is extremely elevated. In a 2014 study (Figure 3), Portugal demonstrated the highest dependency on private vehicle use among EU member countries, with automobiles accounting for 89,9% of total passenger transport (Eurostat, 2017).

With excessive car dependency, cities face increasing challenges due to the elevated number of private vehicles and their associated negative externalities. Frequent car use

leads to increased air pollution and CO₂ emissions but also exacerbates traffic congestion, stress levels, health issues, unsustainable economic results, and environmental problems (European Environment Agency, 2023a, 2023b; Nieuwenhuijsen et al., 2019). Furthermore, compared to PT users, car users face a higher accident risk (Ning et al., 2016). In fact, Portugal has one of the highest mortality rates in road accidents per million inhabitants within the EU (Eurostat, 2022). Overall, higher levels of automobile dependency in urban areas severely impact our societies, causing environmental, social, and economic problems (Cunha et al., 2018; Fu, 2022; Nieuwenhuijsen et al., 2019).

Despite these negative externalities, car dependency has revealed itself a difficult challenge to overcome. It appears that car owners' reluctance to reduce daily usage is influenced by factors such as **comfort, flexibility, privacy, social status, and reduced travel time** (Bergstad et al., 2011). In other words, cars represent more than a simple mode of transportation. They are associated with elevated social status, as well as a source of pleasure, freedom, and independence (Steg, 2005, 2007; Tertoolen et al., 1998). On the other hand, some authors point to a turning moment in the symbolism of the car (Bayart et al., 2020; McDonald, 2015; Zhu & Fan, 2018). According to these researchers, cars are progressively less associated with powerful social status and are increasingly associated with negative experiences, especially by younger city residents (Bayart et al., 2020). The literature reveals that younger generations are reducing car use and changing their overall mobility patterns (McDonald, 2015).

To reach climate neutrality by 2050 successfully, there is a unanimous agreement, on a European level, on the importance of reducing car dependency and developing sustainable transport networks (European Environment Agency, 2023a, 2023b). According to the Encyclopedia of Sustainability in Higher Education, **sustainable mobility** is defined as the provision of the necessary infrastructure, services, technologies, and information to sustain society's transport needs without sacrificing the social and environmental conditions of future generations (Budnitz, 2019). Quality PT systems play a significant role in promoting sustainable transport options by reducing CO₂ levels, improving air quality, decreasing congestion and accident risk, enhancing accessibility, and lowering pollution levels (Adler & van Ommeren, 2016; Borck, 2019; Cruz & Sarmiento, 2020; Holmgren, 2007; Poliak et al., 2017; Tang & Lo, 2008; Uherek et al., 2010).

To enhance sustainable mobility, cities need competitive and eco-friendly transport systems that meet the growing urban demand while simultaneously allowing for a significant reduction in emissions. This can be achieved through improved PT efficiency such as investment in sustainable transport infrastructure; development of Intelligent Transport Systems (ITSs); increased energy efficiency through technological innovation; investment in alternative fuels (e.g., hydrogen, biofuels, synthetic fuels), and well executed urban planning (Banister, 2007; Loukopoulos, 2007; Polat, 2012). For example, transport demand management, such as reducing parking spaces and raising parking fees, is another effective urban planning strategy to reduce car dependency and promote PT usage (Cruz et al., 2017; Inci, 2015). Additionally, the EU has implemented policies to increase PT demand and, decrease car use and ownership. These policies can be differentiated into “Push” and “Pull” measures. “Push” measures refer to any political action taken to restrict the use of private cars (e.g., low emission zones, road closures, and bus priority lanes) (Batty et al., 2015; Fujii et al., 2001). On the contrary, “Pull” measures focus on supporting and enhancing PT services to attract more ridership and break the habit of car use, through initiatives such as fare subsidies and improving PT infrastructure (Batty et al., 2015; Chiaramonti et al., 2021; Gössling & Cohen, 2014). Taxation plays a pivotal role in these strategies, with varying impacts depending on its application since it can be applied as “push” or “pull” measures (Barros et al., 2021; Saleh, 2007). However, so far, modal shift strategies remain below expected levels due to technological, cultural, and regulatory factors that uphold car dependency (Farla et al., 2010). The widespread perception of cars as convenient, flexible, and socially prestigious modes of transport contributes to resistance against modal shift, even in the face of congestion and environmental concerns (Beirão & Cabral, 2007; Tertoolen et al., 1998; Thøgersen, 2009; Vredin Johansson et al., 2006). According to the EU Commission (2017), between 1995 and 2016 PT ridership increased by 18% while car use increased by 23% (EU Commission, 2017).

To combat this tendency further research is required to understand which quality attributes will make PT services attractive/unattractive to the population (Batty et al., 2015). Hence, to reverse the current global emissions trend and reach carbon neutrality in the transport sector, we will require intense political and regulatory action, strategic taxation policies, innovative technological development, infrastructure development, and

citizen involvement (Gärling, 2007; Giles-Corti et al., 2016; Graham-Rowe et al., 2011; Ian Tiseo, 2024; Noblet et al., 2014; Richter et al., 2011; United Nations, 2020).

2.2 Public Transport

PT is the cornerstone of modern urban living, shaped by urban sprawl, population growth, socio-demographic factors, and modern lifestyles (Polat, 2012). PT refers to any transport mode that does not require ownership – “provision of buses, coaches, light and heavy rail, trams and ferries as well as taxis and private hire cars for public use” (Balcombe et al., 2004). Mobility reports refer to public “transport” or “transit” depending on the geographical region. “Public transport” or “collective transport” are widely used in European research, while “mass transit” or “public transit” are terms often used in North America and Southeast Asia (Polat, 2012; Redman et al., 2013). Therefore, for this research, when referring to PT, public/collective “transport” is the selected nomenclature.

PT is pivotal in addressing current societal challenges such as urban sprawl and space constraints, environmental concerns, and the diverse mobility needs of increasingly heterogeneous communities. Therefore, there is a growing urgency, worldwide, to develop good-quality PT (Abenoza et al., 2017). However, on a global scale, due to poor accessibility and inefficient service provision only 50% of urban inhabitants have convenient access to PT (UNSD, 2024). Furthermore, when PT offers poor quality services and passengers don’t have access to a private vehicle, travelers become “captive” to the PT service. A captive PT user is defined as a customer, usually with a low-income status, who is forced to use PT primarily due to its cost-effectiveness, rather than out of preference (Redman et al., 2013).

PT is indispensable for offering a viable alternative to car dependency (Lucas, 2006; Welch & Mishra, 2013), and it can be the key to reducing negative externalities associated with excessive car use (e.g., air pollution, traffic congestion, increased emissions). However, to decrease car dependency PT services must be competitive and attractive to the public (Polat, 2012). Public transport service quality (PT-SQ) directly determines PT’s success (Hamzah et al., 2023). Service quality (SQ) is both a subjective and objective measure as it relies on individual CS and perceived value as well as objective performance metrics – numerical, observable variables (Hauer, 1971; Matas, 2004). When PT provides high-quality service, that either meets or exceeds passengers’ expectations, it becomes competitive and effectively increases PT ridership (Izogo &

Ogba, 2015; Kandampully, 1998). However, achieving and maintaining high-quality service is extremely challenging. PT operators must navigate a complex web of factors, including fluctuating demand patterns, diverse passenger needs, and operational constraints (Polat, 2012).

CS - the subjective assessment of how service delivery meets customers' expectations - is a determining factor of SQ (Abenzo et al., 2017). This measure is often used in transportation to determine PT-SQ (Diana, 2012; Hensher et al., 2003). Furthermore, at the heart of PT-SQ lies the concept of PT's perceived value - the individual assessment of the benefits and detriments of using PT. Perceived value is a determinant factor of PT users' satisfaction and willingness to choose it over other modes of transport (Fu, 2022). Thus, if the perceived value of PT is low, CS is sub-optimal and demand is expected to decrease (Lai & Chen, 2011; van Lierop & El-Geneidy, 2018). On the contrary, if overall quality is elevated, reaching customers' perceived value, PT ridership will increase (Abenzo et al., 2017). Therefore, ensuring SQ and understanding the perceived value for a specific group of customers is imperative for the success of any PT service (Friman et al., 2013; Hensher et al., 2003).

According to PT sector research, SQ is influenced by multiple quality attributes. There is such variability in the literature regarding this topic that different studies have reached numerous alternative conclusions regarding the most impactful quality attributes to PT users. The identified quality attributes that determine SQ, according to some of the most impactful research, include (but are not limited to) **reliability, comfort, cost, accessibility, frequency, travel time, speed, safety, and information provision** (Balcombe et al., 2004; Batty et al., 2015; Beirão & Cabral, 2007; L. Cruz et al., 2017; De Oña et al., 2013; Dong et al., 2022; Efthymiou et al., 2018; Felleson & Friman, 2012; Hensher et al., 2003; Ingvardson & Nielsen, 2019; Ishibashi et al., 2024; Lythgoe & Wardman, 2002; Morfoulaki et al., 2010; Mouwen, 2015; Stradling, 2002; Zhang et al., 2019).

When discussing quality attributes, it is imperative to understand that socioeconomic conditions will also influence the ranking of these attributes (Ishibashi et al., 2024). For instance, studies focusing on students' PSQ have different results from those focusing on the general population. Students have specific requirements and place a disproportionately higher value on affordability compared to all other quality attributes

(Batty et al., 2015). Thus, it is crucial to recognize that each PT quality assessment study should be conducted within a context, taking into account the needs and requirements of that specific demand. Furthermore, any interventions to improve overall service performance must be well-planned and account for system disruptions. High disruption levels during intervention stages have led to a decrease in CS, despite having improved system conditions (Friman, 2004).

2.3 Modal Shift

Modal shift is the change (shift) in the transportation system (mode) (Law Insider Inc., 2024). The potential benefits of *modal shift* extend beyond transport choice, offering a multitude of advantages for both society and the environment (HSRG, 2022). For instance, by encouraging behavior change to use other transport options such as walking, cycling, and PT, citizens can improve health parameters through increased physical activity and increased time spent outdoors. Additionally, this mobility shift can alleviate urban congestion and traffic, leading to shorter travel durations for residents (HSRG, 2022). Thus, modal shift strategies have the potential to enhance individual overall well-being and contribute to a better urban environment (Batty et al., 2015; HSRG, 2022; Rissel et al., 2012).

Unfortunately, despite the notable success of modal shifts, governments tend to focus on other strategies to decrease transport emissions. Political agencies can often prioritize investment in green technology to achieve a “post-carbon” society model (Tawiah & Zakari, 2024). For example, in the UK, the present government policy to decarbonize the transport sector is mainly focused on fuel change from diesel/pollutant fuel cars to zero-emission private vehicles. However, this policy perpetuates the focus on private vehicle ownership and usage, rather than encouraging a modal shift to sustainable/collective transport modes (HSRG, 2022). Furthermore, modal shift strategies are affected by multiple complex challenges. The stigma associated with lower social status and lower income levels for PT users can be a strong factor for car use preference in some demographics (Beirão & Cabral, 2007; Şimşekoğlu et al., 2015; UN, 2013). However, stigma is not homogeneous across all transport modes. Research indicates that rail and metro services are generally perceived more favorably than buses, which are often categorized as inconsistent and overcrowded (Batty et al., 2015; Ben-Akiva & Morikawa, 2002; St-Louis et al., 2014; Suman et al., 2018). Furthermore, car ownership significantly

determines car use and, consequently, the success of modal shift interventions. Car owners have been shown to use PT services significantly less than individuals without a personal vehicle. Stable economic conditions often exacerbate car dependency, as higher income levels are correlated with increased car usage (Lebrand & Theophile, 2022). Additionally, it is important to recognize that car users have different motivations and justifications for their transport choices. Some car owners have a strong emotional attachment to their vehicle, and would not easily be swayed to use PT. Other car users are forced to own a private vehicle because either the accessibility to PT is low/nonexistent or the system is too unreliable. For these individuals, the car will remain the main modal choice, despite associated inconveniences such as traffic, lack of parking space, and elevated costs (Beirão & Cabral, 2007). However, if provided with better PT-SQ, these passengers could be convinced by a modal shift strategy to change their transport choice. Moreover, the interplay between external factors, and socioeconomic characteristics will influence the different traveling needs and preferences of each individual and thus influence the success rate of modal shift (Redman et al., 2013). External contexts can be related to traffic conditions, ticket pricing, route density, traveled distance, trip purpose (leisure or work), urban development, and population density. Conversely, socioeconomic characteristics do not depend on external elements but are specific to each individual. These include education, personal experiences, physical fitness, car ownership, unemployment, income, and other demographic information such as age and gender (Noblet et al., 2014; Redman et al., 2013; Urbanek, 2021b).

In resume, good quality and accessible PT networks together with personalized interventions can successfully support modal shift strategies. However, there is an urgent need for investment by public and private institutions to improve PT-SQ and fight the obstacles of car dependency (Barros et al., 2021; Hirave & Sohoni, 2024).

2.4 Quality Attributes that Impact Public Transport Demand

2.4.1 Public Transport Quality Attributes

According to Redman et al. (2013), PT-SQ is determined by multiple quality attributes organized into two main categories: objective/physical and subjective/perceived attributes (Mouwen & Rietveld, 2013). Perceived attributes such as overall SQ, comfort, safety, and crowding effect are determined based on *customer perception*. On the contrary, objective attributes refer to external elements that can be measured

independently of user opinions. These can be demographic and urban patterns of the area, car ownership/access, frequency, reliability, speed, travel time, accessibility, fare, information provision, and more (Balcombe et al., 2004; Barros et al., 2021a; Mouwen & Rietveld, 2013).

Table 1 presents a list of the key PT quality attributes considered for this document, based on information collected from multiple relevant studies.

Table 1. Public Transport (PT) Quality Attributes.

Public Transport Quality Attributes	
Perceived Attributes	Objective Attributes
• Comfort (Beirão & Cabral, 2007; Polat, 2012) • Crowding Effect (Polat, 2012) • Ease of Transfers (Chowdhury et al., 2015) • Safety (Felleson & Friman, 2012; Ingvardson & Nielsen, 2019) • Pro-environmental Behavior (Beirão & Cabral, 2007)	• Frequency (Taylor et al., 2009) • Reliability (Batty et al., 2015; Polat, 2012) • Fare (Polat, 2012; Redman et al., 2013) • Accessibility (Polat, 2012; Redman et al., 2013) • Travel Distance (Polat, 2012) • Travel Time (Beirão & Cabral, 2007; Polat, 2012; Stradling, 2002) • Speed (Redman et al., 2013) • Information Provision (Beirão & Cabral, 2007) • Socioeconomic and Demographic characteristics (Abenoza et al., 2017)

Source: (Lopes, 2024)

2.4.2 Objective Attributes

Frequency

As emphasized by recent research, frequency is a crucial PT-SQ attribute, making it a focal point in improvement endeavors aimed at consistently improving SQ (Ishibashi et